

Lecture 1:-

Discrete Structures.

- 1) Quizes. (15)
- 2) Assignments (5).
- 3) Book "Discrete Mathematics with its Application." 7th Edition.
Kenneth H. Rosen.

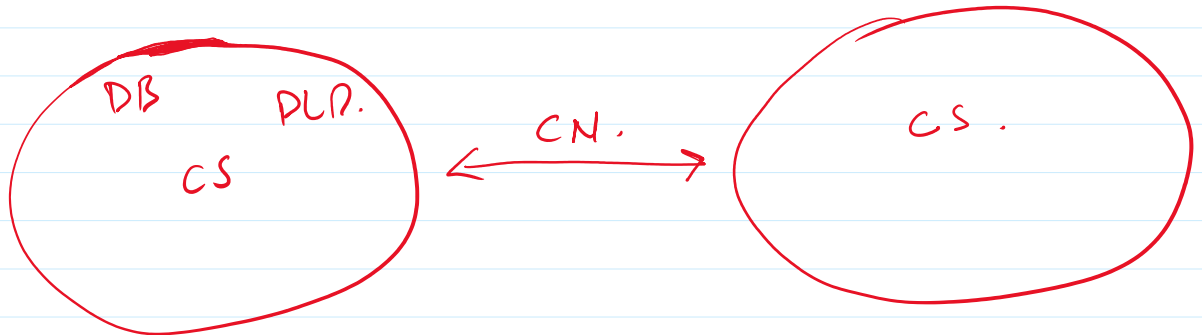
4) Understanding Issue.

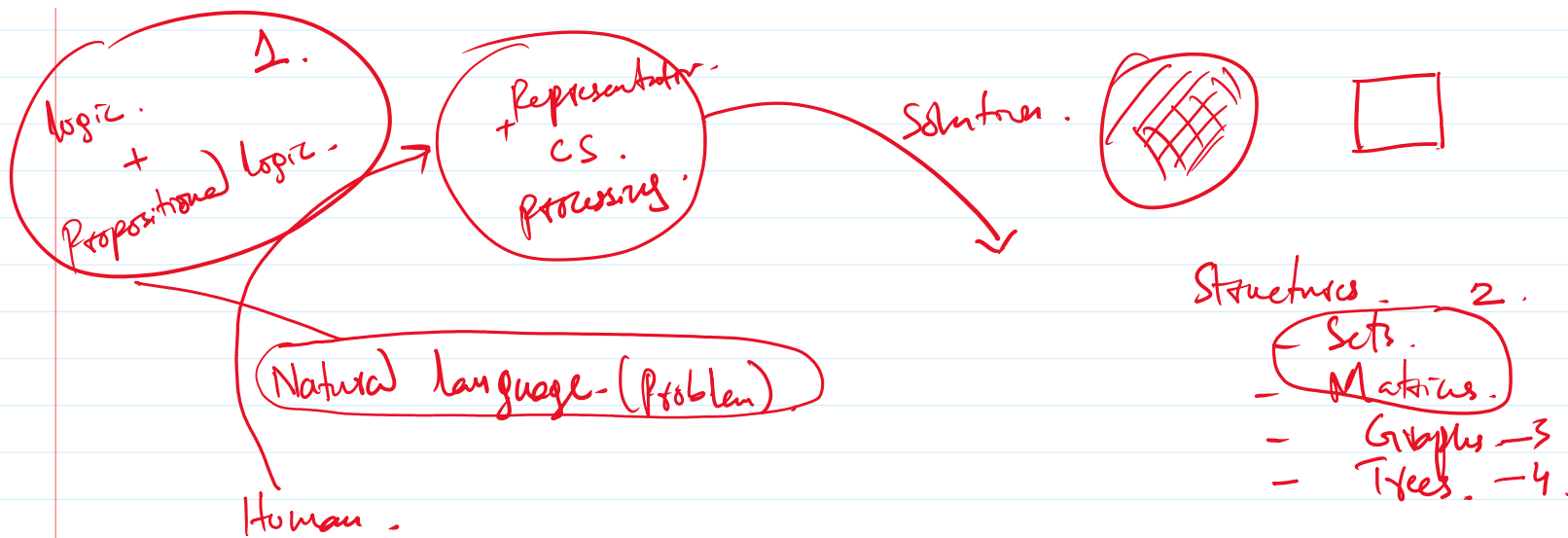
- 4.1) Lecture.
- 4.2) Video Lecture.
- 4.3) Book.
- 4.4) Internet.
- 4.5) Friends.
- 4.6) T.A's.
- 4.7) First 5 minutes of each hr.

Discrete
المنفرد / منفرد

Structures.
ساختار

لغوی بحث
اصطلاحی معنی





Problems in NL Understanding.

(stand) اٹھو، مت بیٹھو (stand)

(sit) اٹھو مت بیٹھو (sit)

Propositional logic.

proposition:- A declarative statement which is either true or false.

Ex 1 :-
P2

$1+1=2$ ✓

Washington is Capital of US. ✓
How are you? X

$2+3=5$ X

Read This Carefully. X.

Propositional Variables:- which hold a proposition.

let $p =$ It is raining.

Operations:-

" . . . "

Negation . $\neg$ → Syntax .

" . . . "

Syntax = لفظی
Semantics = معنی

Operations:

"Not"

Negation: $(\neg)$ → syntax.

Semantics: given values

P	$\neg P$
T	F
F	T

→ Semantics.

Ex 3.
P3

Find

Negation

Today is Friday.

Let p = Today is Friday.

$\neg p$ = Today is not Friday.

not $\times$ = —

Today is not Friday.

= $\boxed{\neg p}$

Let p = Today is Friday.

$\neg(\neg p) = p$

Observation: Operations are not part of propositional variables.

P	$\neg P$	$\neg(\neg P)$
T	F	T
F	T	F